# PHY101 (Mechanics)

Time: 3 hours  Total: 50 marks

## Section A ($10 \times 2 = 20$ marks)

*Choose the single best answer.*

1. A thin uniform rod of length $L$ and mass $M$ spins freely about an axis perpendicular to the rod through a point a distance $a$ from its center. If its angular speed is $\omega$, the angular momentum magnitude about that axis is:

   (a) $M\omega\left(\frac{L^2}{12}+a^2\right)$  (b) $M\omega\left(\frac{L^2}{12}-a^2\right)$
   (c) $M\omega\frac{L^2}{12}$  (d) $M\omega a^2$

2. Two observers $A$ and $B$ are in relative motion with speed $v$. Observer $A$ emits two successive light pulses separated by proper time $\Delta\tau$ on A's clock. Observer $B$ receives the two pulses separated by a time interval:

   (a) $\Delta t = \Delta\tau$
   (b) $\Delta t = \gamma\,\Delta\tau$
   (c) $\Delta t = \gamma K\,\Delta\tau$ where $K = \sqrt{\frac{1+v/c}{1-v/c}}$
   (d) $\Delta t = K\,\Delta\tau$ where $K = \sqrt{\frac{1+v/c}{1-v/c}}$

3. Two identical rigid spheres (radius $R$, mass $M$) roll without slipping down the same incline. Sphere A is solid; sphere B is a thin hollow shell. Their descent times satisfy:

   (a) $t_A < t_B$  (b) $t_A > t_B$  (c) $t_A = t_B$  (d) Cannot decide.

4. A particle moves under a central force $F(r) = -kr\,\hat{\mathbf{r}}$. Which statement is correct?

   (a) All bounded orbits are closed ellipses.
   (b) Angular momentum is not conserved.
   (c) Radial perturbations about circular orbit are simple harmonic with the same frequency as circular motion.
   (d) Energy is not conserved.

5. Consider the force field

$$\vec{F}(x,y,z) = (yz^2 + x^2y)\,\hat{\mathbf{x}} + (xz^2 + xy^2)\,\hat{\mathbf{y}} + (2xyz + z^3)\,\hat{\mathbf{z}}.$$

   Which one of the following statements is correct?

   (a) $\vec{F}$ is conservative everywhere in $\mathbb{R}^3$.
   (b) $\vec{F}$ is conservative only on the plane $z = 0$.
   (c) $\vec{F}$ is not conservative in general.
   (d) $\vec{F}$ is conservative only along curves where $x = y$.

6. Under an inverse-square attractive force, a particle with specific angular momentum $h$ has periapsis $r_p$. The correct energy expression is:

(a) $E = -\frac{\mu}{2r_p}$ (b) $E = \frac{h^2}{2r_p^2} - \frac{\mu}{r_p}$

(c) $E = \frac{h^2}{r_p^2} - \frac{\mu}{2r_p}$ (d) $E = \frac{1}{2}h^2 - \frac{\mu}{r_p}$

7. A mass $m$ sticks to the rim of a disk (mass $M$, radius $R$) after a tangential inelastic collision. Correct conservation principle:

(a) Angular momentum only. (b) Mechanical energy only.
(c) Both. (d) Linear momentum only.

8. Two particles $m_1$ and $m_2$ interact via a central potential; the COM is initially at rest. An external impulse is applied to $m_1$. Then:

(a) COM stays at rest. (b) COM moves. (c) COM stays if impulse radial. (d) Cannot determine.

9. A rigid body rotates about a frictionless pivot. A small attached mass $m$ at distance $r$ detaches instantly with no external torque. Angular momentum about pivot:

(a) Decreases. (b) Increases. (c) Remains unchanged. (d) Undefined.

10. A particle moves in potential $V(r)$ and the effective potential $V_{\text{eff}}$ has an inflection point at $r_0$. Small radial perturbations about $r_0$ produce:

(a) Neutral stability. (b) Stable oscillations. (c) Unstable growth. (d) Forbidden motion.

## Section B ($6 \times 5 = 30$ marks)

*Each question is worth 5 marks. Show reasoning.*

11. An observer finds an object of mass 1 kg moving on a trajectory

$$\vec{r} = \alpha(\beta t^3 - 18t^2)\,\hat{\mathbf{x}} + 2\alpha(\beta t^3 - 18t^2)\,\hat{\mathbf{y}}$$

with constant $\alpha$ and $\beta$ under the influence of a force

$$\vec{F} = \alpha^2(\beta^2 t^2 - 8\beta t + 12)(\hat{\mathbf{x}} + \hat{\mathbf{y}}).$$

Find out if the observer is inertial or non-inertial. Give reason for your answer. (5 marks)

12. A rocket of 300 kg is launched with a speed of 1 km/s vertically from Mohali (latitude $\sim 30°$). The rocket maintains its speed and keeps rising overhead w.r.t. observers on the ground. Obtain the speed of the rocket as seen by local inertial observers instantaneously at rest w.r.t. ground observers, when the rocket has reached a height of 1 km. Find out the magnitude and direction of the Coriolis force acting on the rocket from the perspective of ground observers. (5 marks)

13. Two particles on a frictionless table are located at $(0, 0)$ with mass $m_1 = 1.0\,\text{kg}$ and $(2.0, 0)$ with mass $m_2 = 3.0\,\text{kg}$. A short impulse gives $m_1$ velocity $(4.0\,\hat{\mathbf{x}} + 3.0\,\hat{\mathbf{y}})$ m/s; $m_2$ is untouched.

(a) Find the velocity of the center of mass immediately after the impulse. (2 marks)

(b) Compute the total angular momentum about the origin immediately after the impulse. (3 marks)

14. A particle moves under a central force

$$\vec{F}(r) = -\frac{\mu m}{r^2}\,\hat{\mathbf{r}},$$

where $\mu$ is a positive constant characterising the strength of the attractive inverse-square force. At radius $r = 2r_0$ it has tangential speed $v_t$ such that $h = rv_t = \sqrt{\mu r_0}$ (angular momentum per unit mass).

(a) Write the effective potential per unit mass in terms of $\mu$ and $r_0$. (2 marks)

(b) Determine whether the particle at $r = 2r_0$ is in a bound region. (3 marks)

[illegible] @ $t=0$

15. A source $S$ and receiver $R$ move directly away from each other with constant relative speed $v$. At the instant they pass each other, each emits a flash of light. The Doppler factor is given by $K = \sqrt{\frac{1+v/c}{1-v/c}}$:

(a) Determine the ratio of the arrival times of the two flashes as measured by $R$. (2 marks)

(b) Hence show that $R$ concludes that $S$'s clock runs slow by the Lorentz factor. (3 marks)

16. A rigid body has principal moments of inertia $I_x = 2$, $I_y = 4$, $I_z = 6$ (in arbitrary units). At an instant, its angular velocity is

$$\boldsymbol{\omega} = (1.0,\ 0.2,\ 0.3)$$

expressed in the principal-axis basis.

(a) Compute the kinetic energy of rotation and the angular momentum vector of the rigid body at this instant. (2 marks)

(b) Using only the facts that (i) the motion is torque-free and (ii) both rotational kinetic energy and angular momentum are conserved, describe qualitatively how the direction of $\boldsymbol{\omega}$ can change during the motion. (2 marks)

(c) Explain why $\boldsymbol{\omega}$ cannot remain permanently aligned with the $y$-axis, given the above initial condition. (1 mark)

— **End of paper** —